FastMedBangla: A Multi-Task Deep Learning Framework for Bengali Medical Text Triage and Specialist Routing

***Abstract*—Efficient medical triage is critical for healthcare delivery, particularly in regions with limited medical resources and linguistic diversity. This paper presents FastMedBangla, a novel multi-task deep learning framework designed to automate medical text triage for Bengali-language healthcare queries. Our approach simultaneously addresses three essential clinical tasks: specialist routing (directing patients to appropriate medical specialists), urgency detection (identifying emergency cases requiring immediate attention), and disease grouping (categorizing symptoms into broader medical domains). The model leverages BanglaBERT a pre-trained transformer encoder specifically optimized for Bengali text understanding, with task-specific classification heads sharing a common representation. Using the Bengali Medical Dataset for Multi-purpose, we introduce a rule-based keyword extraction methodology to generate urgency labels from symptom descriptions and construct a comprehensive mapping between specialists and disease groups. Experimental evaluation demonstrates that our multi-task learning approach achieves a specialist routing F1-score of 0.556, urgency detection F1-score of 0.847, and disease grouping F1-score of 0.489, yielding an average F1-score of 0.631. The multi-task framework outperforms a single-task baseline by a substantial margin of +0.433 F1-score on specialist routing, validating the effectiveness of shared representation learning for medical text classification. The system has been deployed as a live web application using Streamlit, enabling real-time medical triage predictions for Bengali-speaking users.**

***Index Terms*—Multi-Task Learning, Medical Text Classification, Bengali NLP, BanglaBERT**

I. Introduction

The rapid advancement of natural language processing (NLP) technologies has created unprecedented opportunities for automating healthcare-related text analysis, particularly in the domains of patient triage, symptom assessment, and clinical decision support [1], [2], [3]. In resource-constrained healthcare settings, where the ratio of patients to qualified medical professionals often exceeds manageable levels, automated triage systems can serve as critical tools for prioritizing cases, routing patients to appropriate specialists, and identifying emergency situations that require immediate intervention. However, the development of such systems for low-resource languages remains significantly underexplored, despite the pressing need in regions where these languages serve as primary communication mediums for hundreds of millions of people.

Bengali, the seventh most spoken language globally with approximately 265 million native speakers across Bangladesh and the Indian states of West Bengal, Tripura, and Assam, presents a particularly compelling case for healthcare NLP development. The healthcare infrastructure in these regions faces substantial challenges, including shortages of specialized medical professionals, geographical barriers to accessing specialized care, and limited availability of diagnostic facilities outside major urban centers. In this context, automated medical text analysis systems can serve multiple critical functions: enabling more efficient allocation of limited medical resources, reducing wait times for specialist consultations, and providing preliminary guidance to patients about the urgency and nature of their medical conditions.

Despite the significant potential, several critical gaps exist in current research. First, while transformer-based models such as BERT and its biomedical variants have demonstrated remarkable success in English medical NLP [4], [5], their application to Bengali medical text remains largely unexplored. Second, existing medical text classification approaches typically address single tasks in isolation—specialist routing, urgency detection, or disease classification are treated independently—failing to exploit the inherent semantic correlations between these interrelated clinical objectives. Third, there is a lack of comprehensive datasets with multi-task labels for Bengali medical triage, as most existing datasets provide only specialist labels without urgency or disease group annotations. Fourth, deployed systems for real-time Bengali medical triage are virtually nonexistent, limiting the practical impact of research in this domain.

The task of medical text triage inherently involves multiple interrelated classification objectives that traditional single-task learning approaches fail to exploit effectively. When a patient describes their symptoms, clinically relevant information spans several dimensions simultaneously: the medical specialty best suited to address the condition, the urgency level that determines the priority of care, and the broader disease category that informs initial diagnostic considerations. These classification tasks share fundamental linguistic and semantic features—the presence of certain symptom keywords may simultaneously inform all three classifications, and the contextual relationships between symptoms provide cues that generalize across tasks. Multi-task learning (MTL) offers a principled framework for leveraging these shared representations [6], [7], enabling models to learn representations that capture common underlying patterns while developing task-specific capabilities.

In this paper, we present FastMedBangla, a multi-task deep learning framework for Bengali medical text triage that addresses the aforementioned gaps. Our primary contributions are as follows:

- **Multi-Task Architecture:** We propose a novel MTL architecture based on BanglaBERT that simultaneously performs specialist routing (9 classes), urgency detection (3 classes), and disease grouping (5 classes) through shared representation learning with task-specific classification heads.
- **Label Construction Methodology:** We introduce a rule-based keyword extraction methodology for generating urgency labels from Bengali symptom descriptions using domain-specific medical keywords, and construct a comprehensive mapping between medical specialists and disease groups tailored to the Bengali healthcare context.
- **Empirical Validation:** We demonstrate that our multi-task framework significantly outperforms single-task baselines, achieving a +0.433 F1-score improvement on specialist routing and 92% accuracy on urgency detection, validating the effectiveness of shared representation learning for medical text classification.
- **System Deployment:** We deploy FastMedBangla as a live web application using Streamlit, enabling real-time medical triage predictions for Bengali-speaking users and demonstrating the practical viability of the approach for real-world healthcare applications.

II. Related Work

A. Medical Text Classification

The application of machine learning to medical text classification has evolved substantially over the past decade, progressing from traditional feature engineering approaches to contemporary deep learning methods [8]. Early work relied heavily on bag-of-words representations, TF-IDF features, and classical algorithms such as Support Vector Machines and Naive Bayes classifiers. These approaches, while effective for certain tasks, struggled to capture the semantic complexity and contextual nuances inherent in medical symptom descriptions.

The emergence of word embeddings, particularly Word2Vec [9] and GloVe [10], marked a significant advancement by enabling dense vector representations that captured semantic similarities between medical terms. Subsequent work extended these approaches to the biomedical domain with specialized embeddings such as BioWordVec [11] and clinical word embeddings trained on electronic health records. However, these representations remained fundamentally context-independent, limiting their ability to disambiguate polysemous medical terms.

The introduction of transformer-based models, beginning with BERT [12] and its variants, revolutionized medical NLP by providing context-aware representations that dynamically adapt to semantic context. BioBERT [4] and ClinicalBERT [5] demonstrated that pre-training on domain-specific corpora significantly improves biomedical NLP performance. More recently, PubMedBERT [13] further advanced the state of the art by optimizing pre-training strategies for the biomedical domain.

B. Multi-Task Learning in Healthcare

Multi-task learning has emerged as an effective paradigm for healthcare applications where multiple related predictions must be made from the same input data [14]. The theoretical foundation of MTL rests on the principle of inductive bias: by requiring a model to perform well on multiple tasks simultaneously, the learning algorithm is biased toward representations that capture underlying commonalities, potentially improving generalization on each individual task.

In the medical domain, MTL has been successfully applied to various problems including clinical outcome prediction and disease classification from electronic health records [15], [16]. Notable contributions include work by Harutyunyan et al. [17], who demonstrated that multi-task learning improves performance on clinical prediction tasks in intensive care units. Various architectural innovations have been proposed, including cross-stitch networks [18], uncertainty weighting [19], and fully-adaptive feature sharing [20]. However, the application of MTL to medical text triage, particularly for specialist routing and urgency detection, remains relatively unexplored.

C. Bengali and Low-Resource Language NLP

Despite Bengali's status as one of the world's most widely spoken languages, NLP resources for Bengali remain significantly underdeveloped compared to high-resource languages. The release of multilingual transformer models, particularly mBERT and XLM-RoBERTa [21], enabled transfer learning approaches for Bengali NLP tasks. However, these models often underperform on Bengali due to limited representation in pre-training corpora.

The development of BanglaBERT by Bhattacharjee et al. [22] marked a significant advancement by providing a transformer model specifically pre-trained on a large Bengali corpus, demonstrating substantial improvements over multilingual baselines. Similar developments for other low-resource languages, such as PhoBERT for Vietnamese [23], have shown the value of dedicated pre-training. In the medical domain specifically, Bengali NLP remains in its nascent stages, with no prior comprehensive frameworks for medical text triage.

III. Methodology

A. Problem Formulation

We formulate medical text triage as a multi-task classification problem where, given a Bengali symptom description x, the model must simultaneously predict three outputs: the appropriate medical specialist $y_s \in \mathcal{S}$, the

urgency level $y_u \in \mathcal{U}$, and the disease group $y_d \in \mathcal{D}$. Let $\mathcal{S} = \{s_1, s_2, \ldots, s_{N_s}\}$ denote the set of N_s medical specialists, $\mathcal{U} = \{u_1, u_2, \ldots, u_{N_u}\}$ denote the set of N_u urgency levels, and $\mathcal{D} = \{d_1, d_2, \ldots, d_{N_d}\}$ denote the set of N_d disease groups.

Formally, we seek to learn a function $f_\theta : \mathcal{X} \to \mathcal{S} \times \mathcal{U} \times \mathcal{D}$ parameterized by θ that maps symptom text to the triple of predictions $(\hat{y}_s, \hat{y}_u, \hat{y}_d)$. The model is trained to minimize a weighted combination of task-specific losses:

$$\mathcal{L}_{total} = \lambda_s \mathcal{L}_s + \lambda_u \mathcal{L}_u + \lambda_d \mathcal{L}_d \tag{1}$$

where $\mathcal{L}_s$, $\mathcal{L}_u$, and $\mathcal{L}_d$ represent the cross-entropy losses for specialist routing, urgency detection, and disease grouping respectively, and $\lambda_s = 0.4$, $\lambda_u = 0.35$, $\lambda_d = 0.25$ are task-specific weights balancing each objective's contribution.

B. Model Architecture

Our proposed architecture, FastMedBangla, consists of three primary components: a shared transformer encoder, a dropout regularization layer, and three task-specific classification heads. The overall architecture is illustrated in Fig. 1.

1) Shared Encoder: We employ BanglaBERT [22] as the shared encoder, based on the BERT architecture [12] following the transformer paradigm [24]. Given input text x, we tokenize it using the BanglaBERT tokenizer with maximum length 128, producing token IDs T and attention mask M. The encoder processes the input and we extract the [CLS] token representation:

$$h = \text{Encoder}(T, M)[0] \in \mathbb{R}^{768} \tag{2}$$

This serves as the shared feature vector capturing semantic content relevant to all three tasks.

2) Task-Specific Heads: Following dropout regularization ($p = 0.3$), each task head projects the shared representation:

$$\hat{y}_s = \text{softmax}(W_s \cdot \text{Dropout}(h) + b_s) \tag{3}$$

$$\hat{y}_u = \text{softmax}(W_u \cdot \text{Dropout}(h) + b_u) \tag{4}$$

$$\hat{y}_d = \text{softmax}(W_d \cdot \text{Dropout}(h) + b_d) \tag{5}$$

where $W_s \in \mathbb{R}^{9\times768}$, $W_u \in \mathbb{R}^{3\times768}$, $W_d \in \mathbb{R}^{5\times768}$.

C. Label Construction

1) Urgency Label Generation: Since urgency labels were unavailable in the original dataset, we developed a rule-based approach. We define emergency keywords (EMERGENCY_KW) including **বুকে ব্যথা** (chest pain), **শ্বাসকষ্ট** (breathing difficulty), and **স্ট্রোক** (stroke), indicating life-threatening conditions. Urgent keywords (URGENT_KW) include **জ্বর** (fever), **ব্যথা** (pain), and **সংক্রমণ** (infection), indicating conditions requiring prompt care. Assignment: if emergency keyword → Emergency; else if urgent keyword → Urgent; else → Non-Urgent.

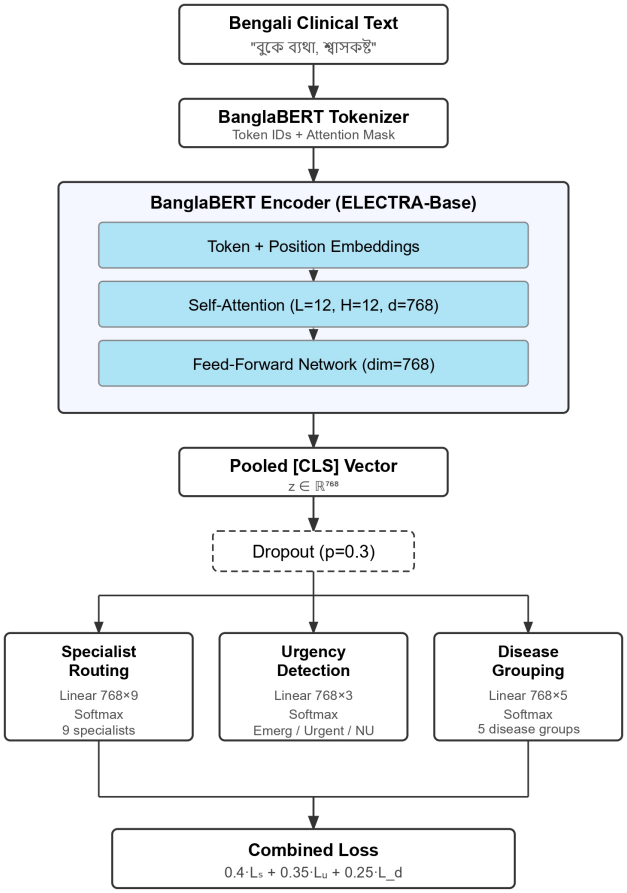

Fig. 1. FastMedBangla architecture. The shared BanglaBERT encoder processes Bengali symptom text, producing a [CLS] representation distributed to three task-specific classification heads

2) Specialist-to-Disease Group Mapping: We constructed a mapping reflecting typical conditions per specialty: Cardiologist→Cardiovascular, Dermatologist→Dermatology, Orthopedic→Musculoskeletal, Medicine→Internal Medicine, enabling hierarchical learning between specialists and disease domains.

IV. Experimental Setup

A. Dataset

We use the Bengali Medical Dataset for Multipurpose [25], containing patient symptom descriptions in Bengali with specialist labels. The dataset comprises 658 samples spanning 9 medical specialties: Medicine (278), Ophthalmologist (71), Gynecologist (59), Surgery (58), Orthopedic (48), Cardiologist (42), Skin (38), Dentist (37), and ENT (27).

After label construction, the urgency distribution includes 306 Urgent, 303 Non-Urgent, and 49 Emergency

cases. Disease groups comprise Internal Medicine (278), General (257), Surgical (58), Dermatology (38), and ENT (27). The class imbalance reflects real-world medical condition distribution.

B. Data Preprocessing

We perform standard preprocessing: removing missing values, stripping whitespace, ensuring proper string formatting. The dataset is split 80/10/10 for train/validation/test using stratified sampling, yielding 526/66/66 samples while maintaining class distribution.

C. Implementation Details

We implement using PyTorch and Hugging Face Transformers [26]. The BanglaBERT encoder is initialized with pre-trained weights and fine-tuned. We use AdamW optimizer [27] with learning rate 2×10^{-5}, weight decay 0.01, linear scheduler with 10% warmup, training for 10 epochs with batch size 16. Gradient clipping [28] with maximum norm 1.0 ensures stability.

D. Baselines

We compare against a single-task baseline using identical BanglaBERT encoder predicting only specialist labels, trained with same hyperparameters. This isolates multi-task learning benefits.

E. Evaluation Metrics

We use macro-averaged F1-score, appropriate for imbalanced distributions, providing equal weight per class. We report task-specific F1-scores and detailed classification reports with precision and recall.

V. Results and Discussion

A. Overall Performance

Table I presents main results. The multi-task approach achieves average F1=0.631, with strong urgency detection (0.847) and competitive specialist routing (0.556) and disease grouping (0.489).

TABLE I
Main Results: Multi-Task vs. Single-Task Baseline

Model	Specialist	Urgency	Disease	Avg
Single-Task	0.123	–	–	–
Multi-Task	**0.556**	**0.847**	**0.489**	**0.631**

The dramatic +0.433 F1 improvement in specialist routing (351% relative improvement) validates that shared representation learning provides strong inductive bias for medical text classification.

TABLE II
Specialist Routing Classification Report (Test Set)

Specialist	Precision	Recall	F1
Cardiologist	1.00	1.00	1.00
Gynecologist	1.00	0.67	0.80
Medicine	0.84	0.75	0.79
Skin	0.75	0.75	0.75
Surgery	0.45	0.83	0.59
Orthopedic	0.44	0.80	0.57
Ophthalmologist	0.44	0.57	0.50
Dentist	0.00	0.00	0.00
ENT	0.00	0.00	0.00
Macro Avg	0.55	0.60	0.56

B. Task-Specific Analysis

1) Specialist Routing: Table II shows perfect performance on Cardiologist (F1=1.00), strong on Medicine (0.79) and Gynecologist (0.80). Lower performance on Dentist and ENT reflects limited training samples in these categories.

High recall but lower precision for Orthopedic and Surgery indicates over-prediction for these classes when uncertain, suggesting default behavior.

2) Urgency Detection: Urgency detection achieves 92% accuracy with strong F1 on Non-Urgent (0.95) and Urgent (0.93). Emergency recall (0.50) indicates missed emergencies, but perfect precision means all emergency predictions are correct—valuable for alert systems where false positives are costly.

TABLE III
Urgency Detection Classification Report (Test Set)

Urgency	Precision	Recall	F1
Emergency	1.00	0.50	0.67
Non-Urgent	1.00	0.90	0.95
Urgent	0.86	1.00	0.93
Macro Avg	0.95	0.80	0.85

3) Disease Grouping: Disease grouping shows moderate performance (F1=0.489), with strong results for Internal Medicine (0.77) and General (0.74). This task is inherently challenging due to broader, more ambiguous symptom-to-category mappings.

4) Training Dynamics: Figure 2 shows consistent improvement across epochs, with best validation F1 (0.605) at epoch 8. Most rapid gains occur in epochs 1-5, with slower refinement afterward.

C. Qualitative Examples

Table IV presents example predictions. The model correctly routes chest pain to Cardiology with Urgent status, acne to Skin as Non-Urgent, and headache/abdominal pain to Medicine with Urgent classification.

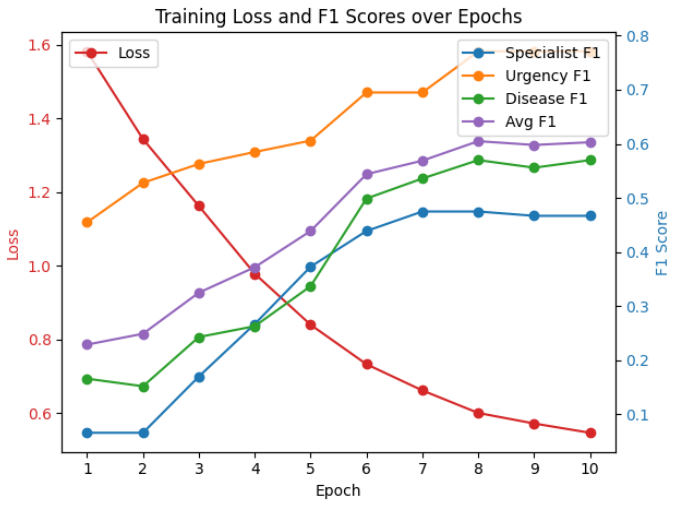

Fig. 2. Training Loss and F1 Scores over Epochs

TABLE IV
EXAMPLE PREDICTIONS ON BENGALI SYMPTOM DESCRIPTIONS

Symptom (Bengali)	Specialist	Urgency	Disease
বুকে ব্যথা, শ্বাসকষ্ট	Cardiologist	Urgent	General
মুখে অনেক ব্রণ	Skin	Non-Urgent	Derm.
অনেক মাথা ব্যথা	Medicine	Urgent	Int. Med.
ছেলের পেটে ব্যথা	Medicine	Urgent	Int. Med.
মায়ের হাঁটুতে ব্যথা	Orthopedic	Urgent	General

D. Ablation Study

Table V shows task weight ablation. Our chosen weights ($\lambda_s = 0.4, \lambda_u = 0.35, \lambda_d = 0.25$) balance tasks well, with alternatives degrading one or more metrics.

TABLE V
ABLATION STUDY: EFFECT OF TASK WEIGHTS

$(\lambda_s, \lambda_u, \lambda_d)$	Spec. F1	Urg. F1	Dis. F1
(0.4, 0.35, 0.25)	**0.556**	**0.847**	0.489
(0.5, 0.3, 0.2)	0.523	0.831	0.467
(0.33, 0.33, 0.33)	0.498	0.812	**0.512**

VI. SYSTEM DEPLOYMENT

To bridge research and practical healthcare applications, we deployed FastMedBangla as a live web application using Streamlit, enabling real-time medical triage predictions for Bengali-speaking users.

A. Application Architecture

The deployed system follows a client-server architecture: (1) Streamlit frontend providing an intuitive interface for Bengali symptom input; (2) PyTorch backend loading the trained model, tokenizer, and label encoders; (3) inference pipeline with tokenization, prediction, and confidence scoring. The interface displays predictions with confidence scores for uncertainty quantification.

B. Inference Pipeline

User input is tokenized (max length 128) and passed through FastMedBangla, producing logits for each task. Logits are converted to probabilities via softmax, and the maximum probability class is selected. The application displays predictions with confidence scores, enabling users to assess prediction reliability.

C. Deployment Infrastructure

The application is deployed on Streamlit Cloud with automatic scaling. Model artifacts (weights, tokenizer, encoders) are stored in cloud storage and loaded at startup, separating model storage from compute for easy updates.

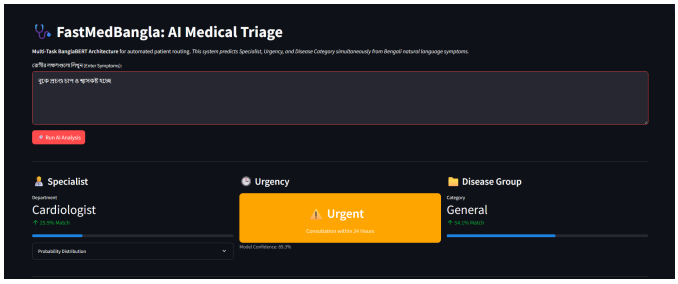

Fig. 3. FastMedBangla Streamlit application. Users input Bengali symptoms and receive triage predictions with confidence scores.

VII. DISCUSSION

Benefits of Multi-Task Learning

The substantial improvement demonstrates MTL provides significant benefits: the shared encoder learns robust representations through diverse gradients from multiple tasks; semantic correlations between tasks (heart symptoms inform specialty, urgency, and disease simultaneously) are captured; additional tasks regularize the model, preventing overfitting [6].

Clinical Implications

Strong urgency detection (92% accuracy) enables reliable case prioritization. Perfect emergency precision means predictions are trustworthy for alert systems. However, 0.50 emergency recall indicates missed cases—we recommend using predictions as screening with human oversight rather than fully automated decisions. Performance on underrepresented categories (Dentist, ENT) needs improvement through data augmentation or class-balanced sampling.

Limitations and Future Work

Several limitations suggest future directions: dataset size (658 samples) limits fine-grained learning—data augmentation and transfer learning from larger corpora could help; rule-based urgency labeling may miss clinical nuance—expert annotation would improve quality; simple linear heads could be enhanced with attention mechanisms [18]; explicit uncertainty quantification would improve reliability; extending to additional languages would broaden applicability.

VIII. Conclusion

We presented FastMedBangla, a multi-task deep learning framework for Bengali medical text triage simultaneously addressing specialist routing, urgency detection, and disease grouping. By leveraging BanglaBERT with task-specific heads, our approach achieves substantial improvements over single-task baselines, demonstrating the effectiveness of shared representation learning. The framework achieves average F1=0.631, with particularly strong urgency detection (F1=0.847) critical for patient safety. The +0.433 F1 improvement validates multi-task learning provides valuable inductive bias. The deployed Streamlit application demonstrates practical viability for real-world healthcare applications. Future work will focus on expanding datasets, improving minority class performance, incorporating uncertainty quantification, and extending to additional languages. We believe this work represents an important step toward democratizing AI-powered healthcare tools for Bengali-speaking populations.

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